

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS
NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 7

HINTS

We will discuss several questions from Chapter 5 and Chapter 6 in Ross' book.

Tutorial Questions

- (1) Two fair dice are rolled. Find
- (i) the joint probability mass function of X and Y , and
 - (ii) the marginal probability mass function of Y , and
 - (iii) the conditional probability mass function of Y given that $X = 4$
- when
- (a) X is the largest value obtained on any die and Y is the sum of the values;
 - (b) X is the value on the first die and Y is the larger of the two values;
 - (c) X is the smallest and Y is the largest value obtained on the dice.

Solution:

- (a) The following table gives the joint pmf and the marginal pmf of Y , where the blank cells mean 0.

$X \backslash Y$	1	2	3	4	5	6	7	8	9	10	11	12
1		$\frac{1}{36}$										
2			$\frac{1}{18}$	$\frac{1}{36}$								
3				$\frac{1}{18}$	$\frac{1}{36}$							
4					$\frac{1}{18}$	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{36}$				
5						$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{9}$	$\frac{1}{18}$		
6							$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{36}$
p_Y	0	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{36}$

One can also write the pmfs as

$$p(x, y) = \begin{cases} \frac{1}{18}, & 2 \leq x \leq 6, x+1 \leq y < 2x; \\ \frac{1}{36}, & 1 \leq x \leq 6, y = 2x; \\ 0, & \text{otherwise} \end{cases}$$

and

$$p_Y(y) = \begin{cases} \frac{y-1}{36}, & 0 \leq y \leq 6; \\ \frac{13-y}{36}, & 7 \leq y \leq 12; \\ 0, & \text{otherwise.} \end{cases}$$

The conditional pmf of Y given $X = 4$ is

$$p_{Y|X}(y|4) = \begin{cases} \frac{2}{7}, & y = 5, 6, 7; \\ \frac{1}{7}, & y = 8; \\ 0, & \text{otherwise.} \end{cases}$$

- (b) The following table gives the joint pmf and the marginal pmf of Y , where the blank cells mean 0.

$X \backslash Y$	1	2	3	4	5	6
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
2		$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
3			$\frac{3}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
4				$\frac{4}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
5					$\frac{5}{36}$	$\frac{1}{36}$
6						$\frac{6}{36}$
p_Y	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

One can also write the pmfs as

$$p(x, y) = \begin{cases} \frac{1}{36}, & 1 \leq x \leq 5, x < y \leq 6; \\ \frac{x}{36}, & 1 \leq x \leq 6, y = x; \\ 0, & \text{otherwise.} \end{cases}$$

and

$$p_Y(y) = \begin{cases} \frac{2i-1}{36}, & y = 1, 2, \dots, 6; \\ 0, & \text{otherwise.} \end{cases}$$

The conditional pmf of Y given $X = 4$ is

$$p_{Y|X}(y|4) = \begin{cases} \frac{2}{3}, & y = 4; \\ \frac{1}{6}, & y = 5, 6; \\ 0, & \text{otherwise.} \end{cases}$$

- (c) The following table gives the joint pmf and the marginal pmf of Y , where the blank cells mean 0.

$X \backslash Y$	1	2	3	4	5	6
1	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$
2		$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$
3			$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$
4				$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{18}$
5					$\frac{1}{36}$	$\frac{1}{18}$
6						$\frac{1}{36}$
p_Y	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

One can also write the joint pmf as

$$p(x, y) = \begin{cases} \frac{1}{18}, & 1 \leq x < y \leq 6; \\ \frac{1}{36}, & 1 \leq x = y \leq 6; \\ 0, & \text{otherwise.} \end{cases}$$

The marginal pmf of Y is the same as that in part (b). The conditional pmf of Y given $X = 4$ is

$$p_{Y|X}(y|4) = \begin{cases} \frac{1}{5}, & y = 4; \\ \frac{2}{5}, & y = 5, 6; \\ 0, & \text{otherwise.} \end{cases}$$

- (2) Consider independent trials, each of which results in outcome i , $i = 0, 1, \dots, k$, with probability p_i , $\sum_{i=0}^k p_i = 1$. Let N denote the number of trials needed to obtain an outcome that is not equal to 0, and let X be that outcome.

- Find $P\{N = n\}$, $n \geq 1$.
- Find $P\{X = j\}$, $j = 1, \dots, k$.
- Show that $P\{N = n, X = j\} = P\{N = n\}P\{X = j\}$.
- Is it intuitive to you that N is independent of X ?

(e) Is it intuitive to you that X is independent of N ?

Solution:

(a) $N \sim \text{Geometric}(1 - p_0)$ and thus $P\{N = n\} = p_0^{n-1}(1 - p_0)$

(b) Let Y be the result of the last trial, then

$$P\{X = j\} = P\{Y = j | Y \neq 0\} = \frac{P\{Y = j\}}{P\{Y \neq 0\}} = \frac{p_j}{1 - p_0}.$$

(c) Using the independence of the trials, we have that

$$P\{N = n, X = j\} = p_0^{n-1}p_j,$$

which is exactly $P\{N = n\}P\{X = j\}$.

(d) Knowing the number of trials (N) reveals no information about which outcome it is in the last trial (X), so N and X are intuitively independent.

(e) Knowing the nonzero outcome in the last trial (X) reveals no information about the number of failed trials, and so X and N are intuitively independent.

(3) Suppose that X and Y are independent geometric random variables with the same parameter p .

(a) Without any computations, what do you think is the value of

$$P\{X = i | X + Y = n\}?$$

Hint: Imagine that you continually flip a coin having probability p of coming up heads. If the second head occurs on the n th flip, what is the probability mass function of the time of the first head?

(b) Verify your conjecture in part (a).

Solution:

(a) Before the second head appears at the n -th position, there are $n - 2$ tails and 1 head, and all such sequences are equally likely. This implies that the conditional distribution of X conditioned on $X + Y = n$ is uniform on $\{1, \dots, n - 1\}$, or,

$$P\{X = i | X + Y = n\} = \frac{1}{n - 1}.$$

(b) Calculate the conditional probability formally:

$$\begin{aligned} P\{X = i | X + Y = n\} &= \frac{P\{X = i, X + Y = n\}}{P\{X + Y = n\}} \\ &= \frac{P\{X = i, Y = n - i\}}{P\{X + Y = n\}} \\ &= \frac{P\{X = i\}P\{Y = n - i\}}{\sum_{k=1}^{n-1} P\{X = k\}P\{Y = n - k\}} \\ &= \frac{(1 - p)^{i-1}p \cdot (1 - p)^{n-i-1}p}{\sum_{k=1}^{n-1} (1 - p)^{k-1}p \cdot (1 - p)^{n-k-1}p} \\ &= \frac{(1 - p)^{n-2}p^2}{\sum_{k=1}^{n-1} (1 - p)^{n-2}p^2} \\ &= \frac{1}{n - 1}. \end{aligned}$$

(4) The joint probability mass function of X and Y is given by

$$\begin{aligned} p(1, 1) &= \frac{1}{8}, & p(1, 2) &= \frac{1}{4} \\ p(2, 1) &= \frac{1}{8}, & p(2, 2) &= \frac{1}{2} \end{aligned}$$

- (a) Compute the conditional mass function of X given $Y = i$, $i = 1, 2$.
- (b) Are X and Y independent?
- (c) Compute $P\{XY \leq 3\}$, $P\{X + Y > 2\}$, $P\{X/Y > 1\}$.

Solution:

- (a) The conditional mass function of X given $Y = 1$ is

$$p_{X|Y=1}(1) = p_{X|Y=1}(2) = \frac{1}{2}$$

and the conditional mass function of X given $Y = 2$ is

$$p_{X|Y=2}(1) = \frac{1}{3}, \quad p_{X|Y=2}(2) = \frac{2}{3}.$$

- (b) X and Y are not independent because the conditional mass function of X given Y is different for $Y = 1$ and $Y = 2$.
- (c)

$$\begin{aligned} P\{XY \leq 3\} &= p(1, 1) + p(1, 2) + p(2, 1) = \frac{1}{2} \\ P\{X + Y > 2\} &= p(1, 2) + p(2, 1) + p(2, 2) = \frac{7}{8} \\ P\{X/Y > 1\} &= p(2, 1) = \frac{1}{8}. \end{aligned}$$

(5) The joint probability density function of X and Y is given by

$$f(x, y) = c(y^2 - x^2)e^{-y}, \quad -y \leq x \leq y, \quad 0 < y < \infty$$

- (a) Find c .
- (b) Find the marginal densities of X and Y .
- (c) Find $\mathbb{E}[X]$.

Solution:

- (a)

$$\begin{aligned} 1 &= \int_0^\infty \int_{-y}^y c(y^2 - x^2)e^{-y} dx dy = c \int_0^\infty \left(y^2 e^{-y}(2y) - e^{-y} \int_{-y}^y x^2 dx \right) dy \\ &= c \int_0^\infty \frac{4}{3} y^3 e^{-y} dy \\ &= c \cdot \frac{4}{3} \cdot 6 = 8c \end{aligned}$$

and hence $c = 1/8$.

- (b) The marginal density of X is

$$\begin{aligned} p_X(x) &= \int_{-\infty}^\infty f(x, y) dy = \int_{|x|}^\infty \frac{1}{8} (y^2 - x^2) e^{-y} dy \\ &= \frac{1}{8} \left(e^{-|x|} (x^2 + 2|x| + 2) - e^{-|x|} x^2 \right) \\ &= \frac{1}{4} e^{-|x|} (|x| + 1), \quad -\infty < x < \infty \end{aligned}$$

and the marginal density of Y is

$$p_Y(y) = \int_{-\infty}^\infty f(x, y) dx = \int_{-y}^y \frac{1}{8} (y^2 - x^2) e^{-y} dx = \frac{1}{6} y^3 e^{-y}, \quad 0 < y < \infty.$$

- (c) Note that $p_X(x) = \frac{1}{4}e^{-|x|}(|x| + 1)$ is an even function, thus $x p_X(x)$ is odd and hence $\mathbb{E}(X) = 0$.
- (6) The joint density function of X and Y is

$$f(x, y) = \begin{cases} xy & 0 < x < 1, 0 < y < 2; \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Are X and Y independent?
 (b) Find the density function of X .
 (c) Find the density function of Y .
 (d) Find $\mathbb{E}(Y)$.
 (e) Find $P\{X + Y < 1\}$.

Solution: (a) Yes. The joint density function factors.

(b) $f_X(x) = x \int_0^2 y dy = 2x$ for $0 < x < 1$;

(c) $f_Y(y) = y \int_0^1 x dx = y/2$ for $0 < y < 2$;

(d) $\mathbb{E}(Y) = \int_0^2 y^2/2 dy = 4/3$;

(e) $P\{X + Y < 1\} = \int_0^1 \int_0^{1-x} xy \, dy dx = \int_0^1 x \frac{1}{2}(1-x)^2 \, dx$
 $= \frac{1}{2} \int_0^1 (x - 2x^2 + x^3) dx = \frac{1}{2} \left[\frac{1}{2}x^2 - \frac{2}{3}x^3 + \frac{1}{4}x^4 \right]_0^1 = \frac{1}{24}.$

Assignment Questions

- (1) Two dice are rolled. Let X and Y denote, respectively, the largest and smallest values obtained. Compute the conditional mass function of Y given $X = i$, for $i = 1, 2, \dots, 6$. Are X and Y independent? Why?

Solution: When $X = i$, there are $2i - 1$ outcomes: $2i - 2$ pairs of (i, j) and (j, i) for $j < i$ and 1 pair of (i, i) . Therefore the conditional mass function is

$$p_{Y|X}(y|X = i) = \begin{cases} 2/(2i - 1), & \text{if } 1 \leq y < i; \\ 1/(2i - 1), & \text{if } y = i; \\ 0, & \text{otherwise.} \end{cases}$$

Since $p_{Y|X}(y|i)$ depends on the value of i and thus X and Y are not independent. Alternatively, one can observe that $X \geq Y$ always holds and thus X and Y are not independent.

- (2) Jay has two jobs to do, one after the other. Each attempt at job i takes one hour and is successful with probability p_i . If $p_1 = 0.3$ and $p_2 = 0.4$, what is the probability that it will take Jay more than 12 hours to be successful on both jobs?

Solution: Let X_i be the number of hours (attempts) spent on job i ($i = 1, 2$). Then each $X_i \sim \text{Geometric}(p_i)$ and X_1, X_2 are independent. The probability in question is

$$\begin{aligned} P\{X_1 + X_2 > 12\} &= P\{X_1 + X_2 \geq 13\} \\ &= \sum_{k=13}^{\infty} P\{X_1 + X_2 = k\} \\ &= \sum_{k=13}^{\infty} \sum_{\ell=1}^{k-1} P\{X_1 = \ell\} P\{X_2 = k - \ell\} \\ &= \sum_{k=13}^{\infty} \sum_{\ell=1}^{k-1} (1 - p_1)^{\ell-1} p_1 \cdot (1 - p_2)^{k-\ell-1} p_2 \\ &= p_1 p_2 \sum_{k=13}^{\infty} \frac{(1 - p_2)^{k-1}}{1 - p_1} \sum_{\ell=1}^{k-1} \left(\frac{1 - p_1}{1 - p_2} \right)^{\ell} \\ &= p_1 p_2 \sum_{k=13}^{\infty} \frac{(1 - p_2)^{k-1}}{1 - p_1} \left(\frac{1 - p_1}{1 - p_2} \right) \frac{1 - \left(\frac{1 - p_1}{1 - p_2} \right)^{k-1}}{1 - \frac{1 - p_1}{1 - p_2}} \\ &= \frac{p_1 p_2}{p_1 - p_2} \sum_{k=13}^{\infty} \left[(1 - p_2)^{k-1} - (1 - p_1)^{k-1} \right] \\ &= \frac{p_1 p_2}{p_1 - p_2} \left(\frac{(1 - p_2)^{12}}{p_2} - \frac{(1 - p_1)^{12}}{p_1} \right) \end{aligned}$$

Plugging in the numbers $p_1 = 0.3, p_2 = 0.4$,

$$P\{X_1 + X_2 > 12\} = 4 \times 0.7^{12} - 3 \times 0.6^{12} = 0.0488 \dots$$

- (3) The joint probability density function of X and Y is given by

$$f(x, y) = \frac{6}{7} \left(x^2 + \frac{xy}{2} \right), \quad 0 < x < 1, 0 < y < 2.$$

- Compute the density function of X .
- Find $P\{X > Y\}$.
- Find $P\{Y > 1/2 | X < 1/2\}$.
- Find $\mathbb{E}(X)$.

(e) Find $\mathbb{E}(Y)$.

Solution:

(a) The marginal density of X is

$$p_X(x) = \int_0^2 f(x, y) dy = \frac{6}{7}(2x^2 + x), \quad x \in (0, 1).$$

(b)

$$\begin{aligned} P\{X > Y\} &= \int_0^1 \int_0^x f(x, y) dy dx \\ &= \int_0^1 \int_0^x \frac{6}{7} \left(x^2 + \frac{xy}{2} \right) dy dx \\ &= \int_0^1 \frac{6}{7} \cdot \frac{5x^3}{4} dx \\ &= \frac{15}{56}. \end{aligned}$$

(c)

$$\begin{aligned} P\{Y > 1/2 | X < 1/2\} &= \frac{\Pr\{Y > 1/2, X < 1/2\}}{\Pr\{X < 1/2\}} \\ &= \frac{\int_{1/2}^2 \int_0^{1/2} f(x, y) dx dy}{\int_0^{1/2} p_X(x) dx} \\ &= \frac{\int_{1/2}^2 \left(\frac{1}{28} + \frac{3}{56}y \right) dy}{\frac{5}{28}} \\ &= \frac{69}{80}. \end{aligned}$$

(d)

$$\mathbb{E}(X) = \int_0^1 x p_X(x) dx = \frac{6}{7} \left(\frac{1}{2} + \frac{1}{3} \right) = \frac{5}{7}.$$

(e) The marginal density of Y is

$$p_Y(y) = \int_0^1 f(x, y) dx = \frac{6}{7} \left(\frac{1}{3} + \frac{1}{4}y \right), \quad 0 < y < 2.$$

Thus

$$\mathbb{E}(Y) = \int_0^2 y p_Y(y) dy = \frac{8}{7}.$$

(4) If X_1 and X_2 are independent exponential random variables with respective parameters λ_1 and λ_2 , find the distribution of $Z = X_1/X_2$. Also compute $P\{X_1 < X_2\}$.

Solution: We shall first find the CDF of Z . It is clear that $Z \geq 0$.

$$\begin{aligned} P(Z \leq z) &= P(X_1 \leq zX_2) = \int_0^\infty \int_0^{zy} \lambda_1 e^{-\lambda_1 x} \lambda_2 e^{-\lambda_2 y} dx dy \\ &= \int_0^\infty \lambda_2 e^{-\lambda_2 y} (1 - e^{-\lambda_1 zy}) dy \\ &= 1 - \frac{\lambda_2}{\lambda_1 z + \lambda_2}. \end{aligned}$$

Differentiating yields the density function of z ,

$$p_Z(z) = \frac{\lambda_1 \lambda_2}{(\lambda_1 z + \lambda_2)^2}, \quad z \geq 0.$$

Last,

$$P\{X_1 < X_2\} = P\{Z < 1\} = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

(5) The joint density function of X and Y is

$$f(x, y) = \begin{cases} x/5 + cy & 0 < x < 1, 1 < y < 5; \\ 0 & \text{otherwise,} \end{cases}$$

where c is a constant.

- (a) What is the value of c ?
- (b) Are X and Y independent?
- (c) Find $P\{X + Y > 3\}$.

Solution:

- (a) By $1 = \int_0^1 \int_1^5 (x/5 + cy) dy dx = 2/5 + 12c$, which gives $c = 1/20$.
- (b) No, the density does not factor.
- (c)

$$\begin{aligned} P\{X + Y \geq 3\} &= \int_0^1 \int_{3-x}^5 (x/5 + cy) dy dx \\ &= \int_0^1 \left[(2+x)x/5 + 25/40 - (3-x)^2/40 \right] dx \\ &= 1/5 + 1/15 + 5/8 - 19/120 = 11/15. \end{aligned}$$